



PERINTHALMANNA



Varsha Insight: Data-Driven Rainfall Monitoring and Forecasting

Submitted by,

Name : Shahla. P
Reg. No : 1839136
Course : GCEMDAD
Institute : G-TECH Centre of Excellence,
Perinthalmanna

Submitted To,

Harifa Thasnim ,Data science Trainer

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EXECUTIVE SUMMARY

Varsha Insight: Data-Driven Rainfall Monitoring and Forecasting System is a comprehensive data analytics and machine learning project developed to analyze historical rainfall patterns and forecast future rainfall trends across India. The project utilizes monthly, seasonal, and annual rainfall records collected from 36 meteorological subdivisions to support climate monitoring, agricultural planning, disaster preparedness, and water resource management.

The project begins with data preprocessing and exploratory data analysis (EDA) to identify rainfall distributions, seasonal variations, rainfall trends, and regional differences. Missing values were handled appropriately, and various visualization techniques were applied to understand rainfall behavior across different years and subdivisions. Rainfall records were further categorized into Drought, Normal, and Heavy Rainfall conditions to support risk assessment.

Feature engineering techniques such as lag features and rolling averages were developed to capture historical rainfall patterns. Machine learning models including Random Forest and XGBoost were implemented and evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 Score. Comparative analysis showed that XGBoost achieved superior predictive performance and was selected as the final forecasting model.

Using the trained model, future rainfall forecasts were generated for all meteorological subdivisions from 2018 to 2030. A Rainfall Alert System was developed to automatically identify potential drought and heavy rainfall conditions based on historical rainfall statistics. In addition, a Farmer Advisory System was integrated to provide actionable recommendations that support agricultural decision-making under different rainfall scenarios.

To improve accessibility and usability, an interactive Streamlit Web Application was developed for rainfall analysis, forecasting, alert monitoring, and advisory recommendations. Tableau and Power BI dashboards were created to provide interactive visualizations and analytical insights. Furthermore, Hadoop MapReduce and Apache Spark technologies were implemented to demonstrate large-scale rainfall data processing and distributed analytics capabilities.

INTRODUCTION

Background

Rainfall plays a vital role in agriculture, water resource management, climate monitoring, and disaster preparedness. India experiences significant variations in rainfall patterns due to geographical diversity and seasonal monsoon systems. Understanding these rainfall patterns is essential for effective planning and decision-making in sectors that depend heavily on climatic conditions. With the increasing availability of historical weather data and advancements in data analytics and machine learning, rainfall trends can be analyzed more effectively to support future planning and risk management.

Need for the Project

Unpredictable rainfall patterns can significantly affect agricultural productivity, water availability, and environmental sustainability. Traditional rainfall monitoring methods mainly focus on historical observations and often lack predictive capabilities. There is a growing need for intelligent systems that can analyze historical rainfall data, forecast future rainfall trends, identify potential drought and heavy rainfall conditions, and provide actionable recommendations. Such systems can help farmers, policymakers, and environmental agencies make informed decisions and improve preparedness for climate-related challenges.

Scope of the Project

This project focuses on analyzing historical rainfall data from 36 meteorological subdivisions of India and forecasting future rainfall trends using machine learning techniques. The scope includes data preprocessing, exploratory data analysis, rainfall categorization, feature engineering, and model development using Random Forest and XGBoost algorithms. The project also incorporates future rainfall forecasting, rainfall alert generation, and farmer advisory recommendations. Additionally, interactive visualizations are developed using Streamlit, Tableau, and Power BI dashboards, while Hadoop MapReduce and Apache Spark are utilized to demonstrate large-scale rainfall data processing and analytics capabilities.

PROBLEM STATEMENT

Rainfall is a critical factor influencing agriculture, water resource management, disaster preparedness, and environmental sustainability in India. Due to climate variability and changing weather patterns, accurately predicting rainfall has become increasingly challenging. Uncertain rainfall conditions can result in droughts, floods, crop losses, and inefficient resource utilization. Therefore, there is a need for an intelligent data-driven system that can analyze historical rainfall patterns, forecast future rainfall trends, detect abnormal rainfall conditions, and provide actionable recommendations to support effective decision-making for farmers, planners, and policymakers.

OBJECTIVES

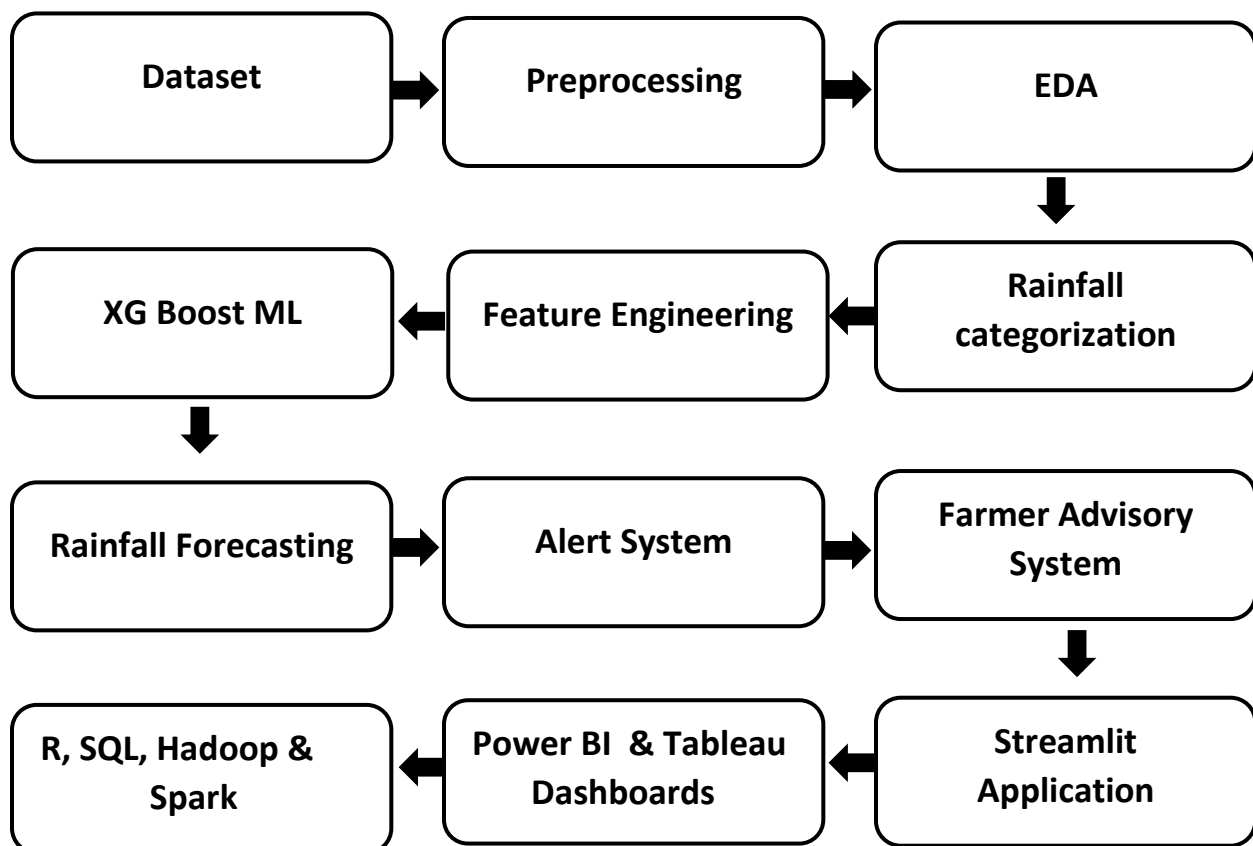
- To analyze historical rainfall data collected from different meteorological subdivisions of India.
- To perform exploratory data analysis and identify rainfall patterns, trends, and seasonal variations.
- To preprocess rainfall data and create meaningful features for forecasting.
- To develop and evaluate Random Forest and XGBoost models for rainfall prediction.
- To compare forecasting models using MAE, RMSE, MAPE, and R^2 Score.
- To forecast future rainfall trends for Indian meteorological subdivisions from 2018 to 2030.
- To develop a rainfall alert system for identifying drought and heavy rainfall conditions.
- To provide farmer advisory recommendations based on forecasted rainfall conditions.
- To develop an interactive Streamlit web application for rainfall monitoring and forecasting.
- To create Tableau and Power BI dashboards for rainfall visualization and decision support.
- To perform rainfall data analysis using Hadoop MapReduce and Apache Spark technologies.
- To support data-driven decision-making in agriculture, water resource management, and disaster preparedness.

PROJECT WORKFLOW

The project begins with collecting historical rainfall data from Indian meteorological subdivisions. The dataset is preprocessed by handling missing values and preparing the data for analysis. Exploratory Data Analysis (EDA) is then performed to identify rainfall patterns, trends, and seasonal variations.

Feature engineering techniques such as lag features and rolling averages are applied to improve forecasting performance. Random Forest and XGBoost models are developed and evaluated, and the best-performing model is selected for future rainfall prediction.

Using the final model, rainfall forecasts are generated for the period 2018–2030. Based on forecasted rainfall conditions, a Rainfall Alert System and Farmer Advisory System are developed to support decision-making. Finally, the results are presented through Streamlit, Tableau, and Power BI dashboards, while Hadoop MapReduce and Apache Spark are used for large-scale rainfall data analysis.



DATASET DESCRIPTION

Dataset Overview

The dataset used in this project contains historical rainfall records collected from 36 meteorological subdivisions of India from 1901 to 2017. It includes monthly, seasonal, and annual rainfall measurements that are used for rainfall analysis, forecasting, and climate trend identification.

Attribute	Value
Source	IMD
Period	1901–2017
Records	4188
Subdivisions	36

Column Description

COLUMN NAME	DESCRIPTION
SUBDIVISION	meteorological subdivision/region in India
YEAR	Year of rainfall observation
JAN	Rainfall received during January (mm)
FEB	Rainfall received during February (mm)
MAR	Rainfall received during March (mm)
APR	Rainfall received during April (mm)
MAY	Rainfall received during May (mm)
JUN	Rainfall received during June (mm)
JUL	Rainfall received during July (mm)
AUG	Rainfall received during August (mm)
SEP	Rainfall received during September (mm)
OCT	Rainfall received during October (mm)
NOV	Rainfall received during November (mm)
DEC	Rainfall received during December (mm)
ANNUAL	Total annual rainfall received during the year (mm)
JF	Combined rainfall of January + February
MAM	Combined rainfall of March + April + May
JJAS	Combined rainfall of June + July + August + September (Southwest Monsoon Season)
OND	Combined rainfall of October + November + December (Northeast Monsoon Season)

METHODOLOGY

The methodology adopted in this project includes data preprocessing, exploratory data analysis, rainfall forecasting, and decision-support system development. Machine learning models were used to predict future rainfall trends, while visualization and big data tools were employed to present and analyze the results. The overall workflow is illustrated through the subsequent methodology stages.

Data Preprocessing

Data preprocessing is an important step in the machine learning pipeline. It helps improve data quality and ensures that the dataset is suitable for analysis and forecasting. In this project, preprocessing involved loading the rainfall dataset, examining its structure, handling missing values, checking duplicate records, and preparing the data for further analysis.

Steps Performed

- Imported the required Python libraries for data analysis and visualization.
- Loaded the rainfall dataset into a Pandas DataFrame.
- Examined dataset dimensions, column names, and data types.
- Identified missing values in rainfall attributes.
- Replaced missing numerical values using mean imputation.
- Checked for duplicate records and ensured data consistency.
- Verified that the cleaned dataset contained no missing values before proceeding to analysis.

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

import warnings
warnings.filterwarnings("ignore")

# Load dataset
df = pd.read_csv(r"../dataset/Sub_Division_IMD_2017.csv")
df.head()
```


	SUBDIVISION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JF	MAM	JJAS	OND
0	Andaman & Nicobar Islands	1901	49.2	87.1	29.2	2.3	528.8	517.5	365.1	481.1	332.6	388.5	558.2	33.6	3373.2	136.3	560.3	1696.3	980.3
1	Andaman & Nicobar Islands	1902	0.0	159.8	12.2	0.0	446.1	537.1	228.9	753.7	666.2	197.2	359.0	160.5	3520.7	159.8	458.3	2185.9	716.7
2	Andaman & Nicobar Islands	1903	12.7	144.0	0.0	1.0	235.1	479.9	728.4	326.7	339.0	181.2	284.4	225.0	2957.4	156.7	236.1	1874.0	690.6
3	Andaman & Nicobar Islands	1904	9.4	14.7	0.0	202.4	304.5	495.1	502.0	160.1	820.4	222.2	308.7	40.1	3079.6	24.1	506.9	1977.6	571.0
4	Andaman & Nicobar Islands	1905	1.3	0.0	3.3	26.9	279.5	628.7	368.7	330.5	297.0	260.7	25.4	344.7	2566.7	1.3	309.7	1624.9	630.8

```
# Basic dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4188 entries, 0 to 4187
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   SUBDIVISION            4188 non-null   object
1   YEAR                   4188 non-null   int64
2   JAN                    4184 non-null   float64
3   FEB                    4185 non-null   float64
4   MAR                    4182 non-null   float64
5   APR                    4184 non-null   float64
6   MAY                    4185 non-null   float64
7   JUN                    4183 non-null   float64
8   JUL                    4181 non-null   float64
9   AUG                    4184 non-null   float64
10  SEP                    4182 non-null   float64
11  OCT                    4181 non-null   float64
12  NOV                    4177 non-null   float64
13  DEC                    4178 non-null   float64
14  ANNUAL                 4162 non-null   float64
15  JF                     4182 non-null   float64
16  MAM                    4179 non-null   float64
17  JJAS                   4178 non-null   float64
18  OND                    4175 non-null   float64
dtypes: float64(17), int64(1), object(1)
memory usage: 621.8+ KB
```

```
df.isnull().sum()
```

```
SUBDIVISION    0
YEAR           0
JAN            4
FEB            3
MAR            6
APR            4
MAY            3
JUN            5
JUL            7
AUG            4
SEP            6
OCT            7
NOV           11
DEC           10
ANNUAL        26
JF            6
MAM           9
JJAS          10
OND           13
dtype: int64
```

```
numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns

for col in numeric_cols:
    df[col].fillna(df[col].mean(), inplace=True)
```

```
df.isnull().sum()
```

```
SUBDIVISION    0
YEAR           0
JAN            0
FEB            0
MAR            0
APR            0
MAY            0
JUN            0
JUL            0
AUG            0
SEP            0
OCT            0
NOV            0
DEC            0
ANNUAL         0
JF             0
MAM            0
JJAS           0
OND            0
dtype: int64
```

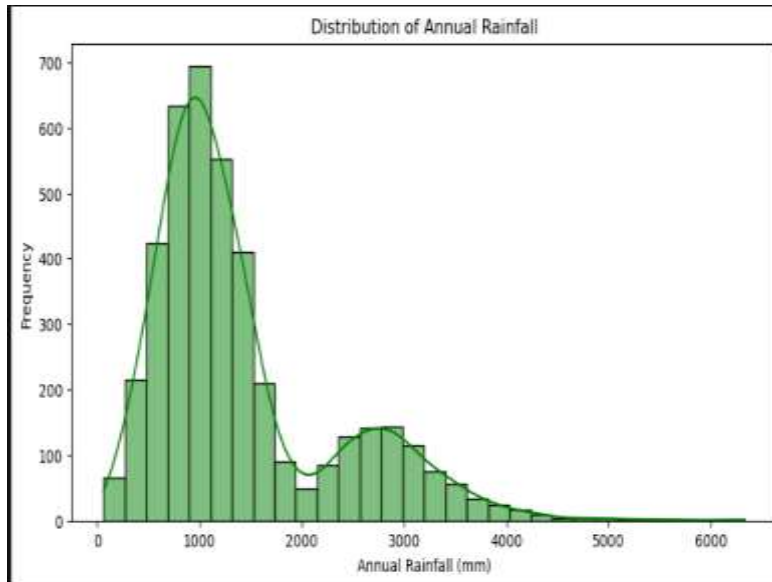
```
df.duplicated().sum()
np.int64(0)
```

After preprocessing, the dataset was cleaned and prepared for exploratory data analysis and rainfall forecasting. Missing values were successfully handled, duplicate records were verified, and the dataset was ready for subsequent modeling tasks.

Exploratory Data Analysis(EDA)

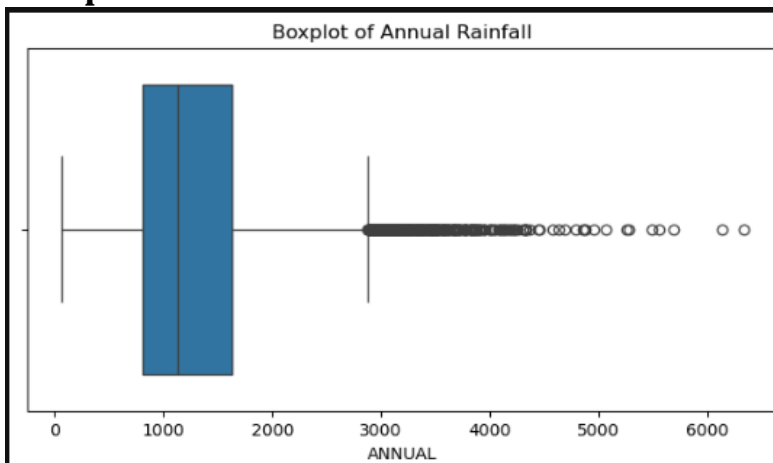
Exploratory Data Analysis was performed to understand rainfall distributions, identify regional rainfall variations, and analyze long-term rainfall trends. Various visualizations were used to gain insights into rainfall behavior across India.

Annual Rainfall Distribution



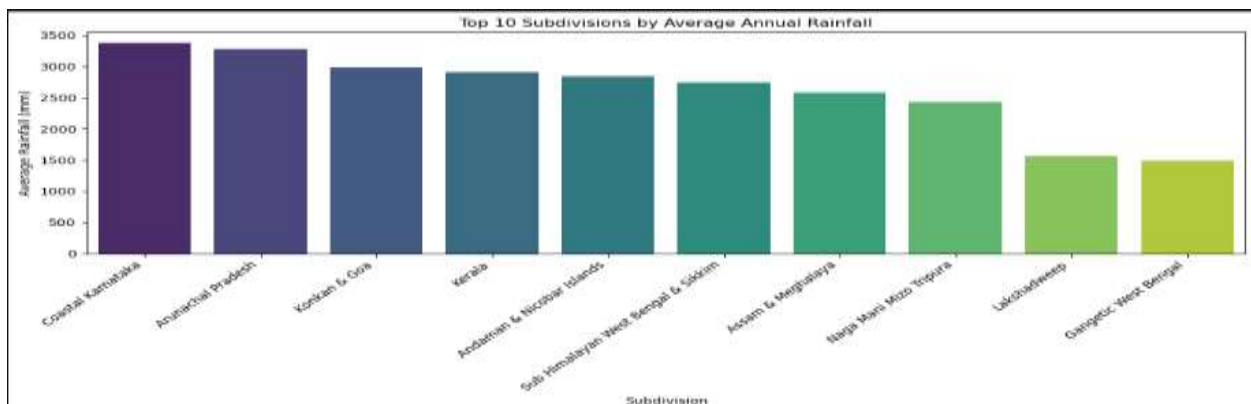
The histogram shows that most annual rainfall observations are concentrated between moderate rainfall ranges, while a smaller number of observations experience very high rainfall. This indicates that rainfall distribution is positively skewed with a few extreme rainfall events.

Boxplot of Annual Rainfall



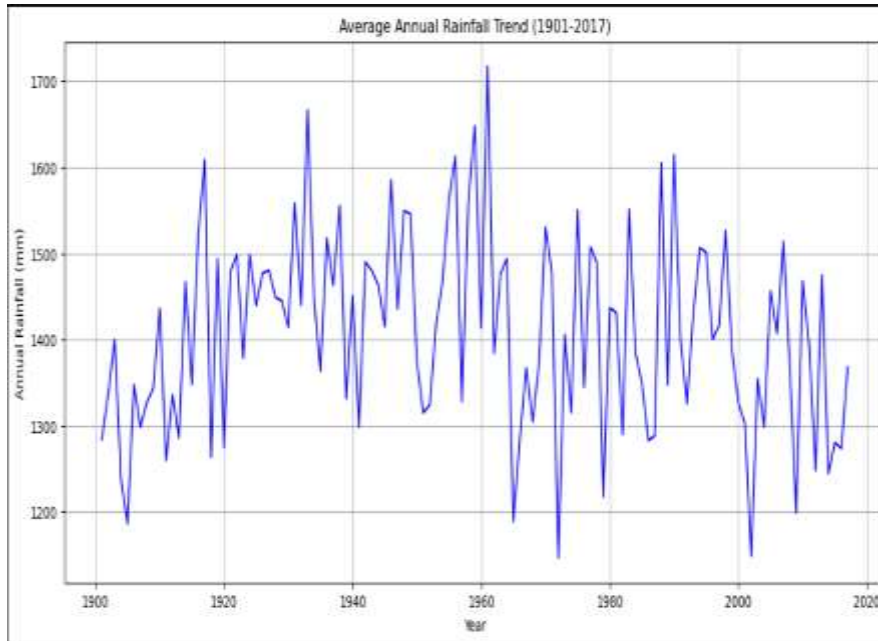
The boxplot highlights the spread of annual rainfall values and identifies several outliers with exceptionally high rainfall. These outliers correspond to regions that receive significantly higher rainfall compared to the national average.

Top 10 Highest Rainfall Subdivisions



The analysis shows that Coastal Karnataka, Arunachal Pradesh, and Konkan & Goa are among the highest rainfall-receiving subdivisions. These regions experience heavy precipitation due to monsoon influences and geographical factors.

Average Annual Rainfall Trend (1901-2017)



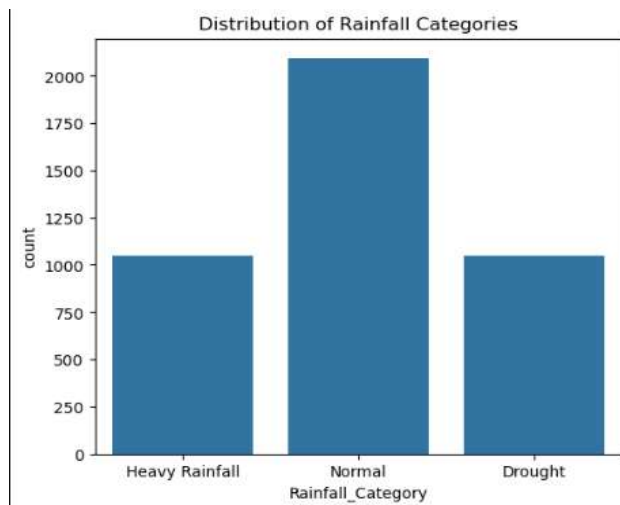
The rainfall trend fluctuates considerably across years without showing a consistent increasing or decreasing pattern. This indicates the presence of rainfall variability, making long-term rainfall forecasting an important task.

Rainfall Categorization

Rainfall records were classified into three categories based on annual rainfall values:

- **Drought** – Low rainfall conditions
- **Normal** – Moderate rainfall conditions
- **Heavy Rainfall** – High rainfall conditions

The categorization was performed using quartile-based thresholds (Q1 and Q3) calculated from annual rainfall data.



Most rainfall records belong to the **Normal** category, while the remaining records are classified as **Drought** and **Heavy Rainfall**. This categorization was later used for rainfall alerts and farmer advisory generation.

Feature Engineering

To improve rainfall forecasting performance, historical rainfall-based features were created.

Features Created

Lag_1 – Rainfall value from the previous year

Lag_2 – Rainfall value from two years before

Lag_3 – Rainfall value from three years before

Rolling_3 – Average rainfall of the previous 3 years

Rolling_5 – Average rainfall of the previous 5 years

```
export the dataset
df = df.sort_values(by=['SUBDIVISION', 'YEAR'])

df['Lag_1'] = df.groupby('SUBDIVISION')['ANNUAL'].shift(1)
df['Lag_2'] = df.groupby('SUBDIVISION')['ANNUAL'].shift(2)
df['Lag_3'] = df.groupby('SUBDIVISION')['ANNUAL'].shift(3)

df['Rolling_3'] = df.groupby('SUBDIVISION')['ANNUAL'].shift(1).rolling(window=3).mean()
df['Rolling_5'] = df.groupby('SUBDIVISION')['ANNUAL'].shift(1).rolling(window=5).mean()

df.dropna(inplace=True)

df.shape

(4000, 25)

df.head()
```

DIVISION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	...	JF	MAM	JIAS	OND	Rainfall_Category	Lag_1	Lag_2	Lag_3	Rolling_3	Rolling_5
idaman & Nicobar Islands	1906	36.6	0.0	0.0	0.0	556.1	733.3	247.7	320.5	...	36.6	556.1	1465.8	475.9	Heavy Rainfall	2566.7	3079.6	2957.4	2867.900000	3099.52
idaman & Nicobar Islands	1907	110.7	0.0	113.3	21.6	616.3	305.2	443.9	377.6	...	110.7	751.2	1327.1	1156.9	Heavy Rainfall	2534.4	2566.7	3079.6	2726.900000	2931.76
idaman & Nicobar Islands	1908	20.9	85.1	0.0	29.0	562.0	693.6	401.4	699.9	...	106.0	591.0	2303.7	575.7	Heavy Rainfall	3347.9	2534.4	2566.7	2816.333333	2897.20
idaman & Nicobar Islands	1910	26.6	22.7	206.3	89.3	224.5	472.7	264.3	337.4	...	49.3	520.1	1701.0	629.0	Heavy Rainfall	3576.4	3347.9	2534.4	3152.900000	3021.00
idaman & Nicobar Islands	1911	0.0	8.4	0.0	122.5	327.3	849.0	253.0	187.1	...	8.4	498.8	1553.6	675.4	Heavy Rainfall	2899.4	3576.4	3347.9	3274.566667	2984.96

25 columns

Machine Learning Model Development

Machine learning models were developed to predict annual rainfall using historical rainfall features. Random Forest and XGBoost algorithms were trained and evaluated using MAE, RMSE, R² Score, and MAPE metrics.

RandomForest Model

```
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

mae = mean_absolute_error(y_test, y_pred_rf)
rmse = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2 = r2_score(y_test, y_pred_rf)
mape = np.mean(
    np.abs((y_test - y_pred_rf) / y_test)
) * 100

print("MAE :", mae)
print("RMSE :", rmse)
print("R² :", r2)
print("MAPE :", mape)
```

MAE : 46.51727540237584
RMSE : 127.3803739868101
R² : 0.9804321194785136
MAPE : 3.210115282117112

The Random Forest model achieved high prediction accuracy with an R² score of 0.9804 and a MAPE of 3.21%, indicating strong forecasting performance.

XGBoost Model

```
mae = mean_absolute_error(y_test, y_pred_xgb)

rmse = np.sqrt(mean_squared_error(y_test, y_pred_xgb))

r2 = r2_score(y_test, y_pred_xgb)

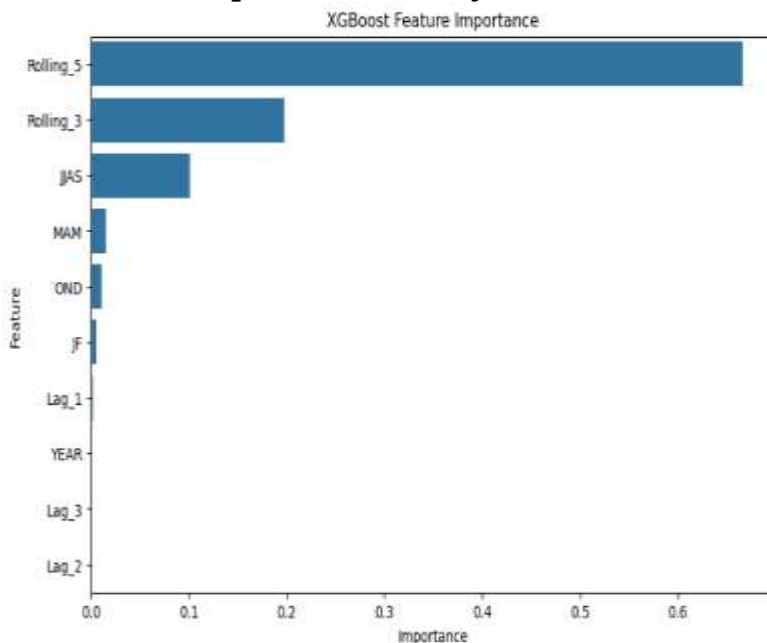
mape = np.mean(
    np.abs((y_test - y_pred_xgb) / y_test)
) * 100

print("MAE :", mae)
print("RMSE :", rmse)
print("R2 :", r2)
print("MAPE :", mape)

MAE : 33.66764452643342
RMSE : 102.95932840475179
R2 : 0.9872159002459393
MAPE : 2.2928152018882755
```

The XGBoost model outperformed Random Forest by achieving a higher R^2 score of 0.9872 and a lower MAPE of 2.29%. Therefore, XGBoost was selected as the final forecasting model.

Feature Importance Analysis



The feature importance analysis indicates that **Rolling_5** and **Rolling_3** are the most influential features in rainfall prediction. This suggests that recent historical rainfall trends play a significant role in forecasting future rainfall. Seasonal rainfall feature **JJAS** also contributed to the model, while other features had comparatively lower importance.

Model Evaluation and comparison

The XGBoost Estimation model achieved the highest R^2 score (**0.9872**) and the lowest MAPE (**2.29%**), indicating superior prediction performance compared to Random Forest. Therefore, XGBoost was selected as the final model for rainfall forecasting. The forecasting model achieved a lower R^2 score because future rainfall values are estimated using historical patterns.

	Model	R2 Score	MAPE (%)
0	Random Forest	0.9804	3.21
1	XGBoost Estimation	0.9872	2.29
2	XGBoost Forecast	0.8706	16.46

Future Rainfall Forecasting

After selecting XGBoost as the final model, future rainfall forecasting was performed for all 36 meteorological subdivisions from **2018 to 2030**. Historical rainfall patterns, lag features, and rolling averages were used to generate future rainfall estimates.

```
mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)

mape = np.mean(np.abs((y_test - y_pred)/y_test))*100

print("MAE :", mae)
print("RMSE :", rmse)
print("R2 :", r2)
print("MAPE :", mape)

MAE : 260.95087563425153
RMSE : 369.55621629599915
R2 : 0.809793752534133
MAPE : 25.011516305115993
```

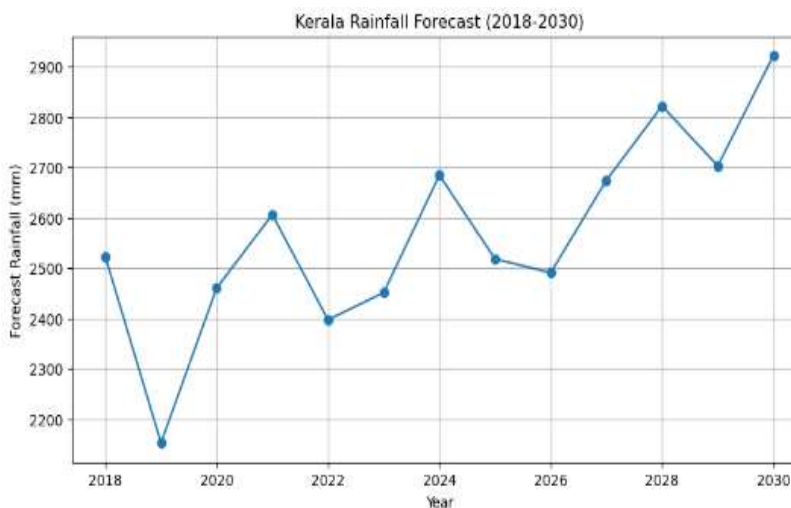
The forecasting model achieved an **R² score of 0.8098** and a **MAPE of 25.01%**. Although the accuracy is lower than the estimation models, it provides a realistic prediction of future rainfall using only historical rainfall information.

Forecast Output

	SUBDIVISION	YEAR	FORECAST_RAINFALL
0	Andaman & Nicobar Islands	2018	3026.937256
1	Andaman & Nicobar Islands	2019	3044.760498
2	Andaman & Nicobar Islands	2020	3094.262451
3	Andaman & Nicobar Islands	2021	3335.996582
4	Andaman & Nicobar Islands	2022	3121.244873

Future rainfall predictions were generated for all meteorological subdivisions from **2018 to 2030**, creating a region-wise rainfall forecast dataset.

Kerala Rainfall Forecast (2018–2030)



The forecast indicates fluctuations in rainfall across the forecast period. Overall, Kerala is expected to experience varying rainfall levels, highlighting the importance of continuous rainfall monitoring and planning.

A future rainfall forecasting dataset covering the period **2018–2030** was successfully generated and used as the foundation for the Rainfall Alert and Farmer Advisory systems.

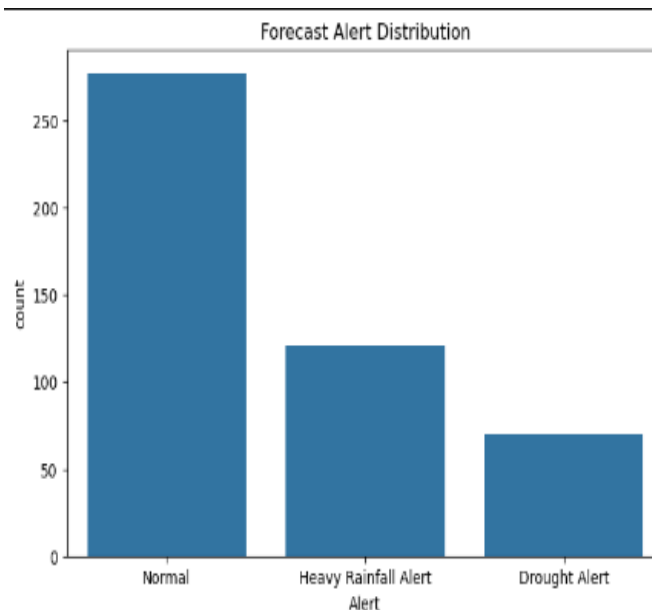
Rainfall Alert System

A Rainfall Alert System was developed to identify potential drought and heavy rainfall conditions using forecasted rainfall values. Historical mean rainfall and standard deviation were used as threshold values for alert generation.

```
forecast_alerts['Alert'].value_counts()
```

```
Alert
Normal          277
Heavy Rainfall Alert  121
Drought Alert    70
Name: count, dtype: int64
```

The alert system generated **277 Normal conditions**, **121 Heavy Rainfall Alerts**, and **70 Drought Alerts** across the forecast period. These alerts help identify regions that may require preventive measures and resource planning.



Farmer Advisory System

	SUBDIVISION	YEAR	FORECAST_RAINFALL	mean	std	Alert	Farmer_Advisory
0	Andaman & Nicobar Islands	2018	3026.937256	2833.221451	527.529461	Normal	Suitable conditions for normal agricultural ac...
1	Andaman & Nicobar Islands	2019	3044.760498	2833.221451	527.529461	Normal	Suitable conditions for normal agricultural ac...
2	Andaman & Nicobar Islands	2020	3094.262451	2833.221451	527.529461	Normal	Suitable conditions for normal agricultural ac...
3	Andaman & Nicobar Islands	2021	3335.996582	2833.221451	527.529461	Normal	Suitable conditions for normal agricultural ac...
4	Andaman & Nicobar Islands	2022	3121.244873	2833.221451	527.529461	Normal	Suitable conditions for normal agricultural ac...

The system automatically generates advisory recommendations based on forecasted rainfall conditions. For drought situations, water conservation and drought-resistant crops are recommended. During heavy rainfall conditions, drainage improvement and flood protection measures are suggested. Normal rainfall conditions are considered suitable for regular agricultural activities.

The Farmer Advisory System successfully transformed rainfall forecasts into actionable recommendations, helping farmers make informed decisions and improve agricultural planning.

STREAMLIT WEB APPLICATION

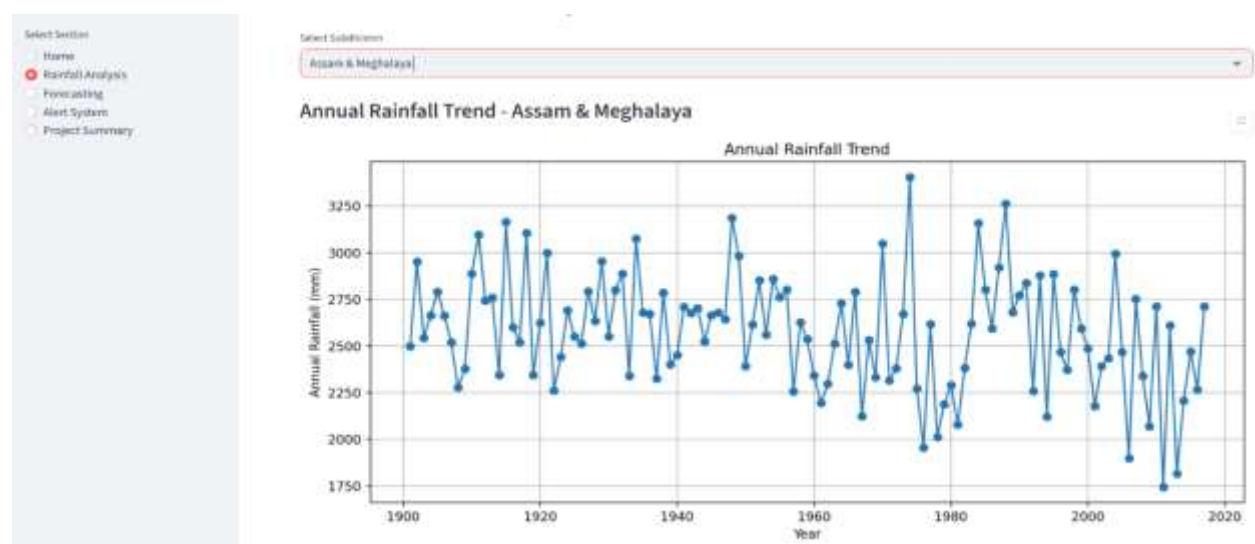
To provide an interactive and user-friendly interface for rainfall analysis and forecasting, a Streamlit web application was developed. The application enables users to explore rainfall trends, visualize forecast results, and access rainfall alerts and advisory information through an intuitive dashboard.

Home page



The Home Page provides an overview of the Varsha Insight system. It displays key dataset information such as total records, number of meteorological subdivisions, forecast period, and a preview of the rainfall dataset. This page serves as the entry point for navigating different modules of the application.

Rainfall Analysis Module



The Rainfall Analysis module enables users to explore historical rainfall trends for individual meteorological subdivisions. Users can select a region and visualize long-term annual rainfall patterns through interactive charts for better climate analysis.

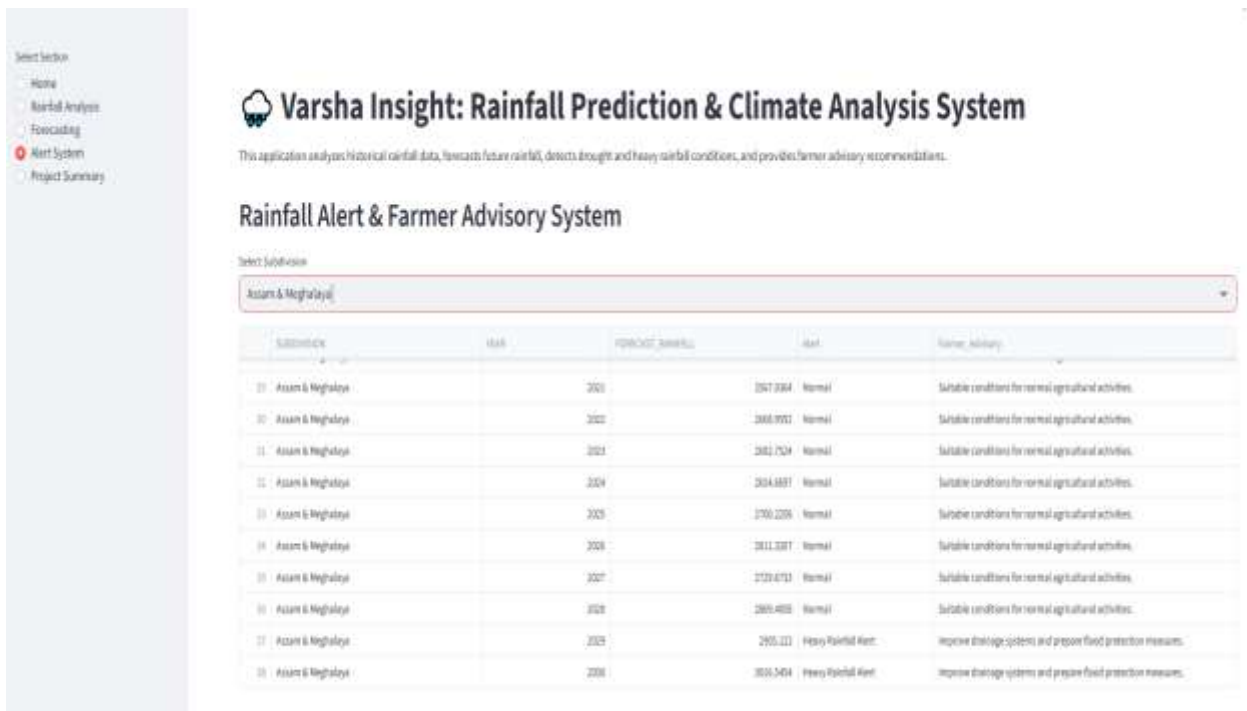
Rainfall Forecasting Module



Rainfall Forecast 2018-2030 - Andaman & Nicobar Islands

This module presents rainfall forecasts generated using the XGBoost forecasting model. Users can select a subdivision and view predicted rainfall values for the period 2018–2030 along with forecast visualizations.

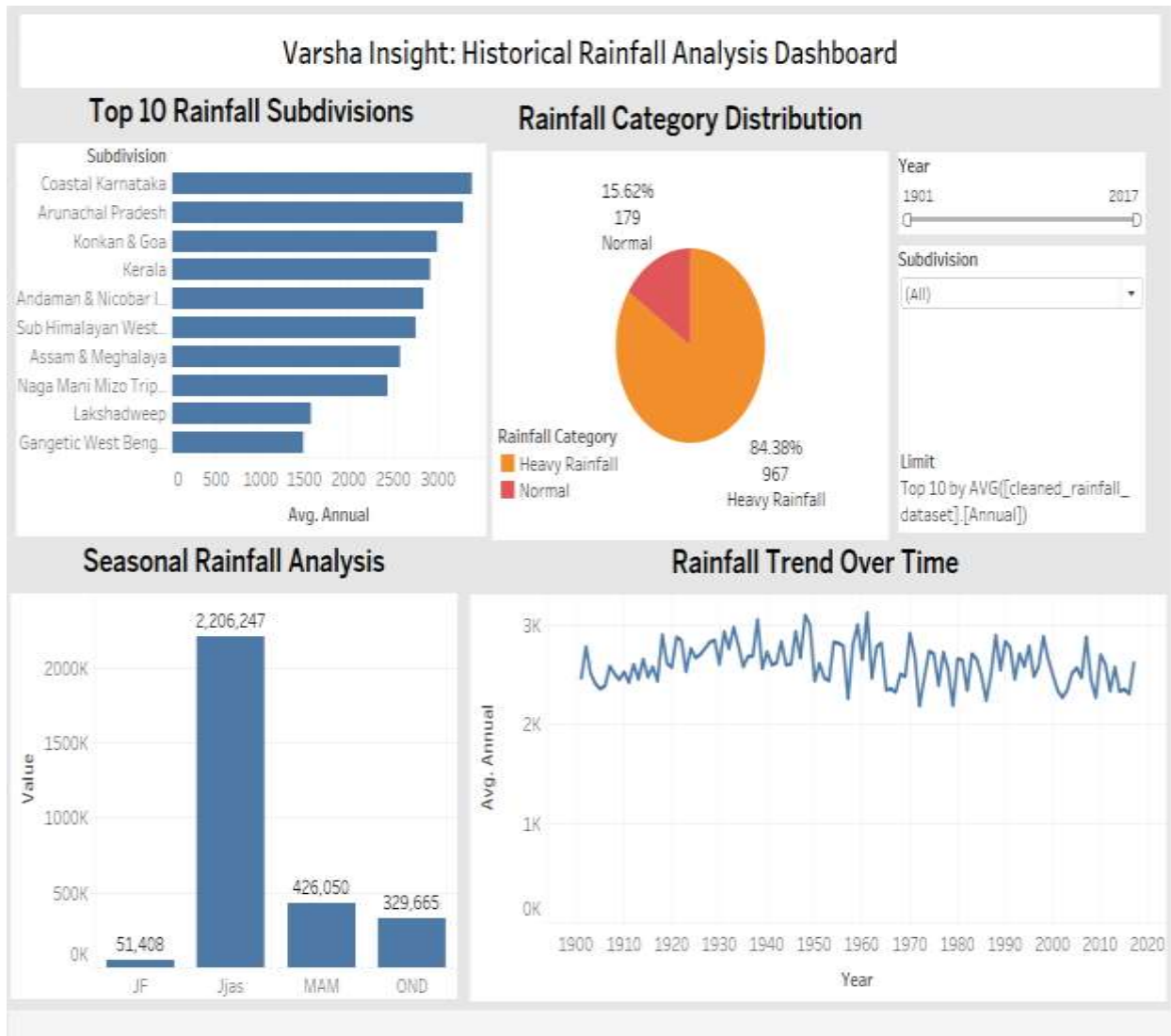
Rainfall Alert and Farmer Advisory Module



The Alert System identifies potential drought and heavy rainfall conditions based on forecasted rainfall values. The Farmer Advisory component provides recommendation messages to support agricultural planning and decision-making.

TABLEAU DASHBOARD

Tableau was used to develop an interactive dashboard for rainfall analysis and visualization. The dashboard provides insights into rainfall distribution, seasonal rainfall patterns, rainfall categories, and long-term rainfall trends across Indian meteorological subdivisions. Interactive filters allow users to explore specific years and regions dynamically.



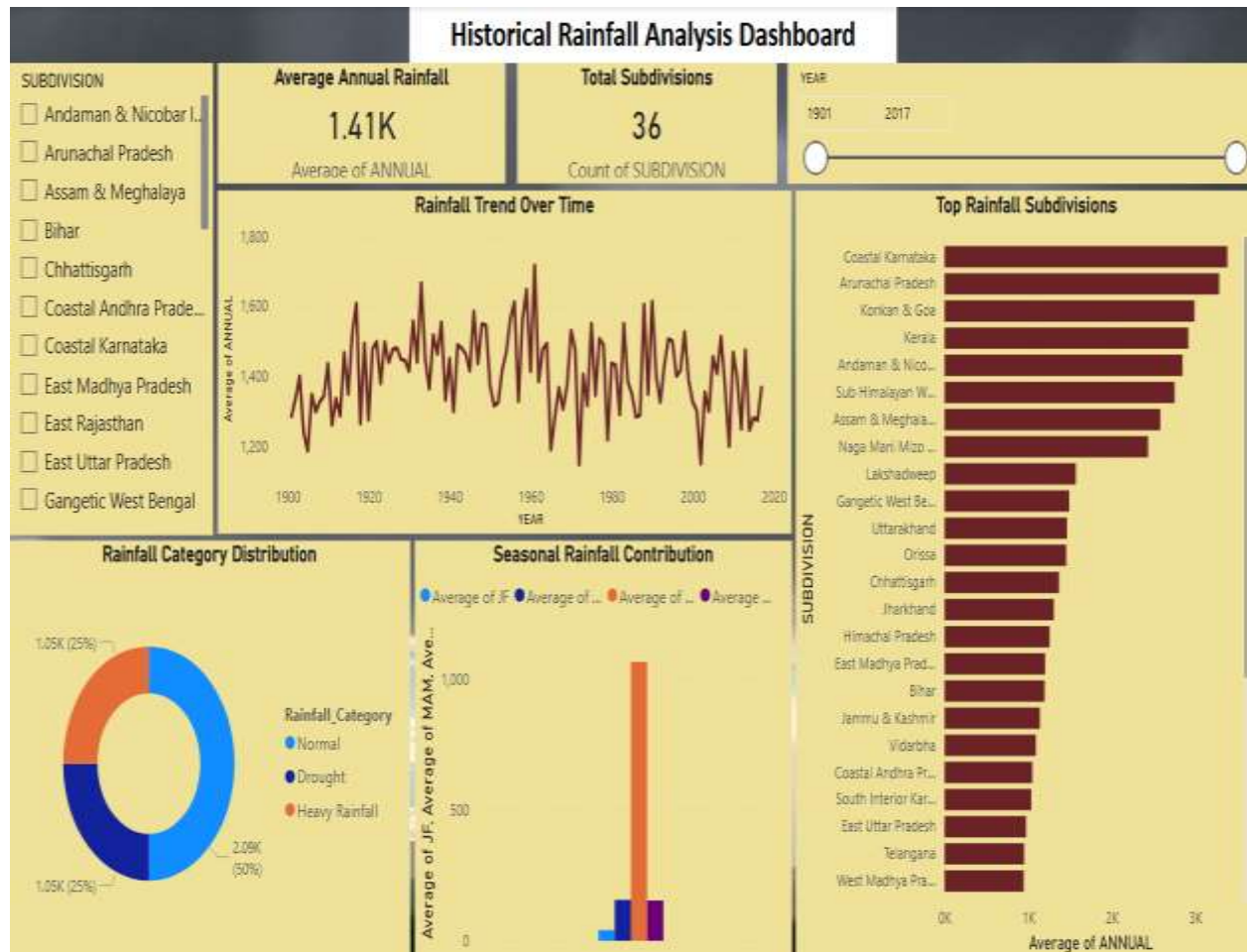
The dashboard highlights the top rainfall-receiving subdivisions in India, with Coastal Karnataka, Arunachal Pradesh, and Konkan & Goa showing the highest average rainfall. Seasonal analysis indicates that the JJAS (Southwest Monsoon) period contributes the largest share of annual rainfall. The rainfall category distribution shows that most records fall under the Heavy Rainfall category. The rainfall trend visualization reveals fluctuations in rainfall over time, demonstrating the variability of climatic conditions across years.

POWER BI DASHBOARD

v

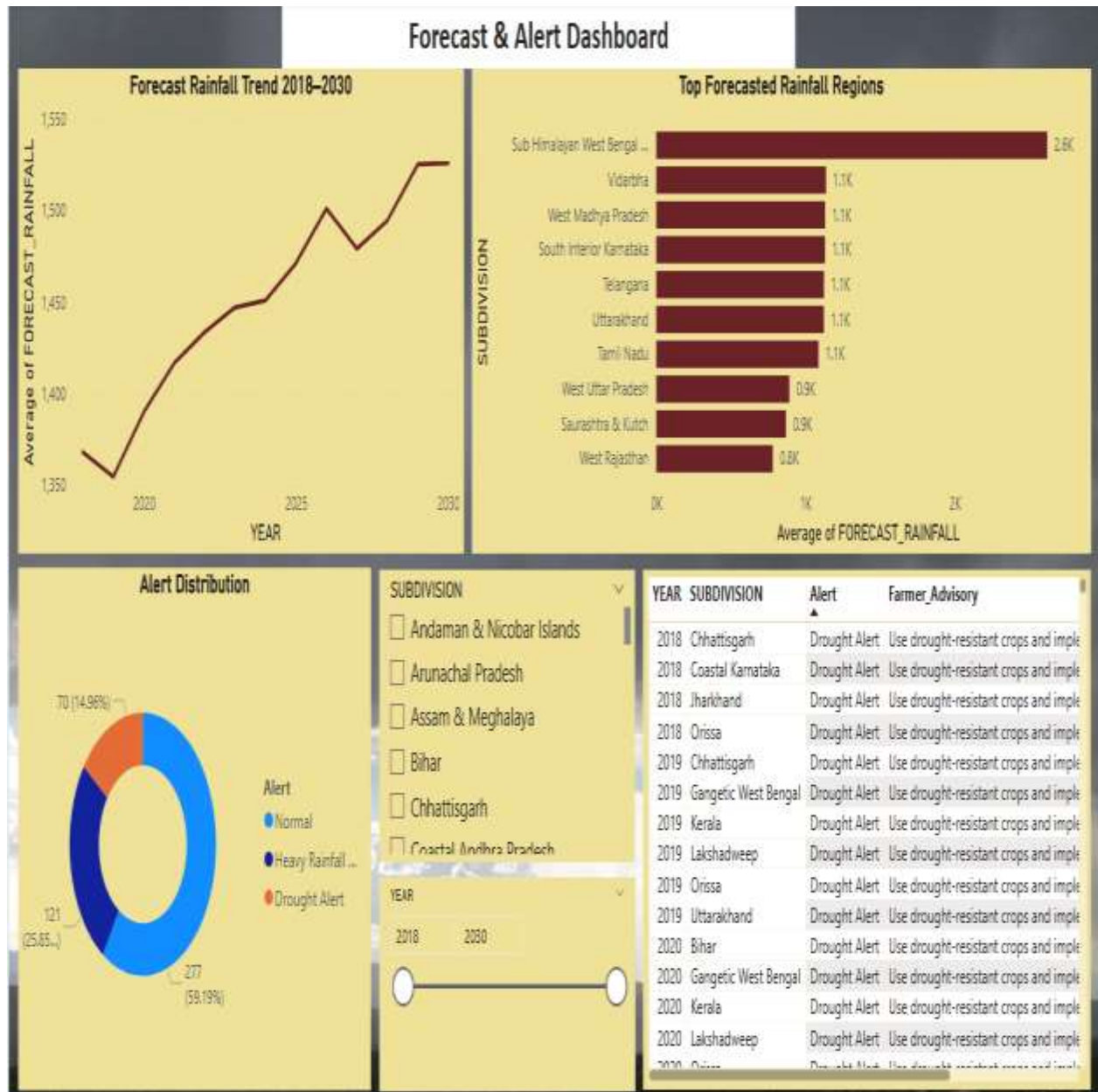
Historical Rainfall Analysis Dashboard

Power BI was used to develop interactive dashboards for rainfall analysis, forecasting, and alert monitoring. The dashboards provide visual insights into historical rainfall patterns, seasonal rainfall distribution, rainfall forecasts, drought alerts, and farmer advisory recommendations. Interactive filters allow users to explore rainfall information across different meteorological subdivisions and years.



The Historical Rainfall Analysis Dashboard provides an overview of long-term rainfall patterns across India. The dashboard displays the average annual rainfall, total number of meteorological subdivisions, rainfall category distribution, seasonal rainfall contribution, and rainfall trends over time. Interactive filters enable users to analyze rainfall data for specific subdivisions and years. The dashboard highlights that Coastal Karnataka, Arunachal Pradesh, and Konkani & Goa receive the highest average rainfall, while the JJAS season contributes the largest share of annual rainfall.

Forecast and Alert Dashboard



The Forecast and Alert Dashboard presents rainfall forecasts generated for the period 2018–2030. It identifies regions with high forecasted rainfall and categorizes forecast results into Normal, Heavy Rainfall Alert, and Drought Alert conditions. The dashboard also integrates Farmer Advisory recommendations to support agricultural planning and risk management. Interactive filters allow users to explore forecast results by subdivision and year, enabling effective monitoring and decision-making.

SQL ANALYSIS

Database Creation and Data Import

```
• CREATE DATABASE rainfall_project;
• USE rainfall_project;
• SHOW TABLES;
• SELECT *
  FROM final_rainfall_dataset;
• SELECT *
  FROM final_rainfall_dataset
  LIMIT 10;
```

	SUBDIVISION	YEAR	JAN	FEB	MAR	APR	MAY
▶	Andaman & Nicobar Islands	1901	49.2	87.1	29.2	2.3	528.1
	Andaman & Nicobar Islands	1902	0	159.8	12.2	0	446.1
	Andaman & Nicobar Islands	1903	12.7	144	0	1	235.1
	Andaman & Nicobar Islands	1904	9.4	14.7	0	202.4	304.1
	Andaman & Nicobar Islands	1905	1.3	0	3.3	26.9	279.1

The cleaned rainfall dataset was imported into a MySQL database named rainfall_project. SQL queries were used to retrieve, summarize, and analyze rainfall information across meteorological subdivisions.

Rainfall Trend Analysis

	SUBDIVISION	Avg_Rainfall
▶	Coastal Karnataka	3380.64
	Arunachal Pradesh	3283.08
	Konkan & Goa	2987.53
	Kerala	2914.25
	Andaman & Nicobar Islands	2845.11
	Sub Himalayan West Bengal & Sikkim	2750.55
	Assam & Meghalaya	2579.13
	Naga Mani Mizo Tripura	2432.72
	Lakshadweep	1570.43
	Gangetic West Bengal	1490.61

This query identifies the top rainfall-receiving subdivisions based on average annual rainfall. Coastal Karnataka, Arunachal Pradesh, Kerala, and Konkan & Goa were among the highest rainfall regions.

Rainfall Status Distribution

	RAINFALL_STATUS	Total_Records
▶	Flood Risk / Extreme Rainfall	939
	Normal Rainfall	406
	Heavy Rainfall Alert	210
	Below Normal Rainfall	526
	Drought Warning	946
	Severe Drought Risk	1161

The query summarizes the number of records belonging to each rainfall category, helping identify drought-prone and flood-prone years.

Advanced SQL Analysis

	SUBDIVISION	Avg_Rainfall	Rainfall_Rank
▶	Coastal Karnataka	3380.64	1
	Arunachal Pradesh	3283.08	2
	Konkan & Goa	2987.53	3
	Kerala	2914.25	4
	Andaman & Nicobar Islands	2845.11	5
	Sub Himalayan West Bengal & Sikkim	2750.55	6
	Assam & Meghalaya	2579.13	7
	Naga Mani Mizo Tripura	2432.72	8
	Lakshadweep	1570.43	9
	Gangetic West Bengal	1490.61	10
	Uttarakhand	1464.44	11
	Orissa	1455.45	12
	Chhattisgarh	1369.14	13

Window functions were used to rank meteorological subdivisions according to average rainfall.

Average Rainfall by Decade

	SUBDIVISION	Avg_Rainfall	Rainfall_Rank
▶	Coastal Karnataka	3380.64	1
	Arunachal Pradesh	3283.08	2
	Konkan & Goa	2987.53	3
	Kerala	2914.25	4
	Andaman & Nicobar Islands	2845.11	5
	Sub Himalayan West Bengal & Sikkim	2750.55	6
	Assam & Meghalaya	2579.13	7
	Naga Mani Mizo Tripura	2432.72	8
	Lakshadweep	1570.43	9
	Gangetic West Bengal	1490.61	10
	Uttarakhand	1464.44	11
	Orissa	1455.45	12
	Chhattisgarh	1369.14	13

This query evaluates long-term rainfall trends by calculating average rainfall for each decade.

Key Findings

- SQL efficiently analyzed rainfall records stored in the database.
- Coastal Karnataka and Arunachal Pradesh recorded the highest average rainfall.
- Rainfall categories revealed both drought and heavy rainfall years.
- Decadal analysis showed variations in rainfall patterns over time.
- Window functions helped rank subdivisions based on rainfall intensity.

R PROGRAMMING ANALYSIS

R Programming was used for statistical analysis, data visualization, machine learning model development, and rainfall forecasting. The rainfall dataset was loaded into R and analyzed using packages such as **dplyr**, **ggplot2**, and **xgboost**. The analysis helped identify rainfall trends, visualize regional patterns, and generate future rainfall forecasts.

Data Loading and Exploration

```
> library(readr)
> library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
> # Load cleaned dataset
```

```
> df <- read_csv("cleaned_rainfall_dataset.csv")
```

```
Rows: 4188 Columns: 20
```

```
— Column specification —
```

```
Delimiter: ","
```

```
chr (2): SUBDIVISION, Rainfall_Category
```

```
dbl (18): YEAR, JAN, FEB, MAR, APR, MAY, JUN, JUL, AUG, SEP, OCT,
```

```
i Use `spec()` to retrieve the full column specification for this .
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet
```

```
> # View first rows
```

```
> head(df)
```

```
# A tibble: 6 × 20
```

	SUBDIVISION		YEAR	JAN	FEB	MAR	APR	MAY	JUN
	<chr>		<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Andaman & Nicobar Is...		1901	49.2	87.1	29.2	2.3	529.	518.
2	Andaman & Nicobar Is...		1902	0	160.	12.2	0	446.	537.
3	Andaman & Nicobar Is...		1903	12.7	144	0	1	235.	480.
4	Andaman & Nicobar Is...		1904	9.4	14.7	0	202.	304.	495.
5	Andaman & Nicobar Is...		1905	1.3	0	3.3	26.9	280.	629.
6	Andaman & Nicobar Is...		1906	36.6	0	0	0	556.	733.

```
> # Dataset summary
```

```
> summary(df)
```

SUBDIVISION	YEAR	JAN	FEB
Length:4188	Min. :1901	Min. : 0.00	Min. : 0.0
Class :character	1st Qu.:1930	1st Qu.: 0.60	1st Qu.: 0.5
Mode :character	Median :1959	Median : 6.00	Median : 6.5
	Mean :1959	Mean :18.94	Mean :21.6
	3rd Qu.:1988	3rd Qu.:22.00	3rd Qu.:26.6
	Max. :2017	Max. :583.70	Max. :403.5

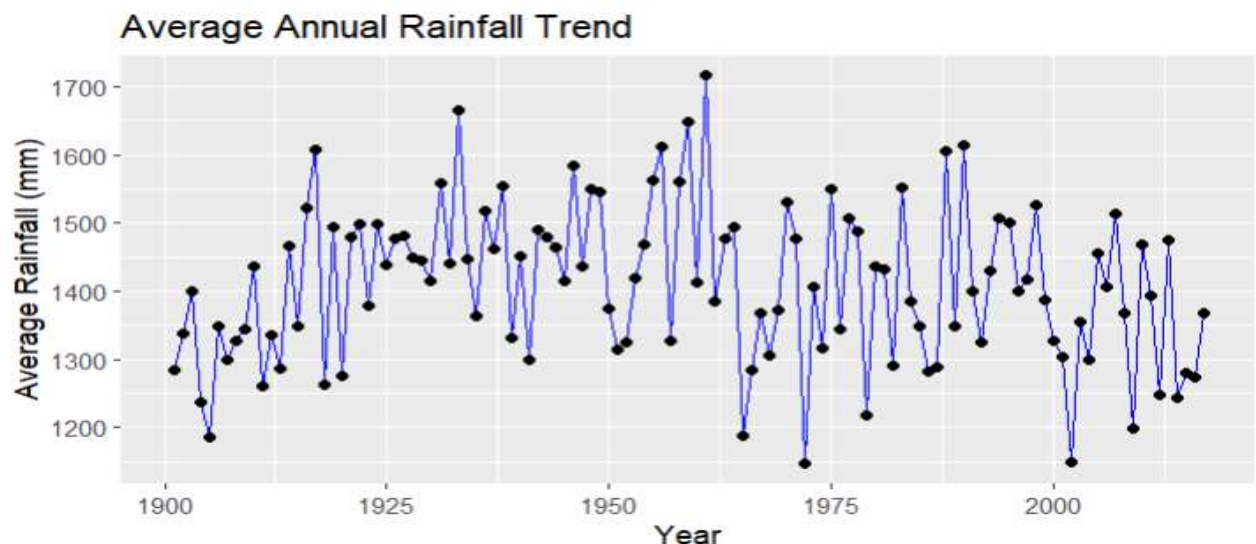
JUN	JUL	AUG	SEP
Min. : 0.40	Min. : 0.0	Min. : 0.0	Min. : 0.0
1st Qu.: 70.88	1st Qu.:175.9	1st Qu.:156.0	1st Qu.:100.4
Median :139.00	Median :285.2	Median :258.9	Median :173.5
Mean :230.12	Mean :347.0	Mean :289.7	Mean :197.3
3rd Qu.:304.30	3rd Qu.:418.4	3rd Qu.:377.5	3rd Qu.:265.8
Max. :1609.90	Max. :2362.8	Max. :1664.6	Max. :1222.6

ANNUAL	JF	MAM	JJAS
Min. : 62.3	Min. : 0.00	Min. : 0.00	Min. : 57.4
1st Qu.: 805.3	1st Qu.: 4.00	1st Qu.:24.18	1st Qu.:574.4
Median :1124.2	Median :19.00	Median :75.15	Median :881.5
Mean :1409.4	Mean :40.52	Mean :155.82	Mean :1063.5
3rd Qu.:1634.1	3rd Qu.:50.10	3rd Qu.:196.90	3rd Qu.:1284.5
Max. :6331.1	Max. :699.50	Max. :1745.80	Max. :4536.5

The cleaned rainfall dataset was imported into R using the readr package. Basic exploratory analysis was performed to examine dataset structure, summary statistics, missing values, total records, subdivision count, and rainfall category distribution.

Data Visualization

Several visualizations were developed using the **ggplot2** package to understand rainfall patterns and trends. These included annual rainfall trends, top rainfall-receiving subdivisions, rainfall category distribution, and Kerala rainfall history



XGBoost Model Development and Forecasting

An XGBoost model was developed in R using lag features and rolling rainfall averages. The model was evaluated using MAE, RMSE, MAPE, and R^2 metrics. After training, future rainfall forecasts were generated for all meteorological subdivisions for the period 2018–2030.

```
> cat("MAE:", round(mae_value, 2), "\n")
MAE: 249.2
> cat("RMSE:", round(rmse_value, 2), "\n")
RMSE: 359.32
> cat("R2:", round(r2_value, 4), "\n")
R2: 0.8202
> cat("MAPE:", round(mape_value, 2), "%\n")
MAPE: 23.44 %
> # Actual vs Predicted plot
> result_df <- data.frame()
```

	SUBDIVISION	YEAR	FORECAST_RAINFALL
1	Andaman & Nicobar Islands	2018	2867.810
2	Andaman & Nicobar Islands	2019	2740.935
3	Andaman & Nicobar Islands	2020	2808.684
4	Andaman & Nicobar Islands	2021	2738.273
5	Andaman & Nicobar Islands	2022	2847.524
6	Andaman & Nicobar Islands	2023	2825.736
7	Andaman & Nicobar Islands	2024	2841.216
8	Andaman & Nicobar Islands	2025	2834.230
9	Andaman & Nicobar Islands	2026	2816.942
10	Andaman & Nicobar Islands	2027	2795.803
11	Andaman & Nicobar Islands	2028	2776.959
12	Andaman & Nicobar Islands	2029	2859.136
13	Andaman & Nicobar Islands	2030	2741.348
14	Arunachal Pradesh	2018	2548.178

Key Findings

- R successfully handled rainfall data loading, exploration, and visualization.
- Historical rainfall trends and regional variations were effectively visualized.
- The XGBoost model demonstrated strong predictive capability using lag and rolling rainfall features.
- Future rainfall forecasts were generated for all subdivisions from 2018 to 2030.
- R served as an effective platform for statistical analysis and machine learning-based rainfall forecasting.

Hadoop MapReduce Implementation

Apache Hadoop MapReduce was used to perform distributed rainfall data processing. The rainfall dataset was stored in HDFS (Hadoop Distributed File System), and Python-based Mapper and Reducer programs were developed using Hadoop Streaming. The objective was to calculate the average annual rainfall for each meteorological subdivision in a distributed environment.

Hadoop Execution

```
C:\Users\HP>start-dfs.cmd
C:\Users\HP>start-yarn.cmd
starting yarn daemons
C:\Users\HP>jps
19424 Jps
17620 DataNode
6756 ResourceManager
7284 NodeManager
18412 NameNode
```

The rainfall dataset was uploaded into HDFS and processed using Hadoop Streaming. The MapReduce job executed successfully, distributing data across mapper and reducer tasks.

```
C:\Users\HP>cd C:\hadoop_rainfall
C:\hadoop_rainfall>hdfs dfs -mkdir /rainfall_project
C:\hadoop_rainfall>hdfs dfs -put C:\Users\HP\Downloads\final_rainfall_dataset.csv /rainfall_project/
C:\hadoop_rainfall>hdfs dfs -ls /rainfall_project
Found 1 items
-rw-r--r-- 3 HP supergroup 557331 2026-06-01 10:09 /rainfall_project/final_rainfall_dataset.csv
C:\hadoop_rainfall>hadoop jar %HADOOP_HOME%\share\hadoop\tools\lib\hadoop-streaming*.jar -D mapreduce.job.
dataset.csv -output /rainfall_output -mapper "C:\Users\HP\AppData\Local\Programs\Python\Python314\python.exe
Programs\Python\Python314\python.exe reducer.py" -file mapper.py -file reducer.py
2026-06-01 10:09:37,892 WARN streaming.StreamJob: -file option is deprecated, please use generic option -f
packageJobJar: [mapper.py, reducer.py, /C:/Users/HP/AppData/Local/Temp/hadoop-unjar2395295852439455478/] [
71493274869.jar tmpDir=null
2026-06-01 10:09:38,514 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /
2026-06-01 10:09:38,655 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /
2026-06-01 10:09:39,060 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
2026-06-01 10:09:39,466 INFO mapred.FileInputFormat: Total input files to process : 1
2026-06-01 10:09:39,597 INFO mapreduce.JobSubmitter: number of splits:2
2026-06-01 10:09:39,707 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1780288679863_0001
2026-06-01 10:09:39,708 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-06-01 10:09:39,880 INFO conf.Configuration: resource-types.xml not found
2026-06-01 10:09:39,880 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-06-01 10:09:40,530 INFO impl.YarnClientImpl: Submitted application application_1780288679863_0001
2026-06-01 10:09:40,565 INFO mapreduce.Job: The url to track the job: http://LAPTOP-2T9BE0FM:8088/proxy/ap
2026-06-01 10:09:40,565 INFO mapreduce.Job: Running job: job_1780288679863_0001
2026-06-01 10:09:40,833 INFO mapreduce.Job: Job job_1780288679863_0001 running in uber mode : false
2026-06-01 10:09:49,836 INFO mapreduce.Job: map 0% reduce 0%
2026-06-01 10:09:54,956 INFO mapreduce.Job: map 100% reduce 0%
2026-06-01 10:10:02,058 INFO mapreduce.Job: map 100% reduce 100%
```

Output Results

Andaman & Nicobar Islands	2845.11
Arunachal Pradesh	3283.08
Assam & Meghalaya	2579.13
Bihar	1196.56
Chhattisgarh	1369.14
Coastal Andhra Pradesh	1050.30
Coastal Karnataka	3380.64
East Madhya Pradesh	1203.04
East Rajasthan	656.50
East Uttar Pradesh	975.86
Gangetic West Bengal	1490.61
Gujarat Region	917.83
Haryana Delhi & Chandigarh	528.44
Himachal Pradesh	1256.78
Jammu & Kashmir	1141.16
Jharkhand	1307.69
Kerala	2914.25
Konkan & Goa	2987.53
Lakshadweep	1570.43
Madhya Maharashtra	881.43
Matathwada	791.75
Naga Mani Mizo Tripura	2432.72
North Interior Karnataka	717.19
Orissa	1455.45
Punjab	591.44
Rayalseema	764.99
Saurashtra & Kutch	496.40
South Interior Karnataka	1037.50
Sub Himalayan West Bengal & Sikkim	2750.55
Tamil Nadu	940.47
Telangana	952.97
Uttarakhand	1464.44
Vidarbha	1093.54
West Madhya Pradesh	948.15
West Rajasthan	294.13
West Uttar Pradesh	822.00

The Hadoop MapReduce analysis generated average annual rainfall values for all 36 meteorological subdivisions. The results confirmed that regions such as Coastal Karnataka, Arunachal Pradesh, Kerala, and Konkan & Goa receive significantly higher rainfall compared to arid regions such as West Rajasthan and Saurashtra & Kutch.

Apache Spark Analysis

Dataset loading and schema validation using PySpark.

Apache Spark was used to perform distributed rainfall data analysis. The rainfall dataset was loaded into a Spark DataFrame, and subdivision-wise average annual rainfall was calculated using Spark's groupBy and aggregation functions. The analysis demonstrates Spark's capability for large-scale data processing.

```
C:\Users\HP\OneDrive\Desktop\varsha_insight_project>pyspark
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

      ____
     / ___/
    / __/
   /___/
  version 4.1.1

Using Python version 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026 16:04:56)
Spark context Web UI available at http://LAPTOP-2T9BE0FM.mshome.net:4040
Spark context available as 'sc' (master = local[*], app id = local-1780290229252).
SparkSession available as 'spark'.
>>> df = spark.read.csv(
...     "C:/Users/HP/OneDrive/Desktop/varsha_insight_project/final_rainfall_dataset.csv",
...     header=True,
...     inferSchema=True
... )
>>> df.show(5)
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|SUBDIVISION|YEAR|JAN|FEB|MAR|APR|MAY|JUN|JUL|AUG|SEP|OCT|NOV|DEC|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Andaman & Nicobar...|1981|49.2|87.1|29.2|2.3|528.8|517.5|365.1|481.1|332.6|388.5|558.2|33.6|
|Andaman & Nicobar...|1982|0.0|159.8|12.2|0.0|446.1|537.1|228.9|753.7|666.2|197.2|359.0|160.5|
|Andaman & Nicobar...|1983|12.7|144.0|0.0|1.0|235.1|479.9|728.4|326.7|339.0|181.2|284.4|225.6|
|Andaman & Nicobar...|1984|9.4|14.7|0.0|202.4|384.5|495.1|502.0|168.1|820.4|222.2|308.7|40.1|
|Andaman & Nicobar...|1985|1.3|0.0|3.3|26.9|279.5|628.7|368.7|330.5|297.0|260.7|25.4|344.7|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

only showing top 5 rows
>>> df.printSchema()
root
 |-- SUBDIVISION: string (nullable = true)
 |-- YEAR: integer (nullable = true)
 |-- JAN: double (nullable = true)
 |-- FEB: double (nullable = true)
 |-- MAR: double (nullable = true)
```

Subdivision-wise average annual rainfall generated using Apache Spark.

	A	B
1	SUBDIVISION	Average_Annual_Rainfall
2	Madhya Maharashtra	881.4316239
3	South Interior Karnataka	1037.555556
4	Kerala	2914.247009
5	East Rajasthan	656.5017094
6	Tamil Nadu	940.4700855
7	Matathwada	791.7452991
8	Chhattisgarh	1369.136752
9	East Madhya Pradesh	1203.035043
10	Lakshadweep	1570.429666
11	Gangetic West Bengal	1490.612821
12	Punjab	591.4367521
13	Konkan & Goa	2987.531624
14	Jammu & Kashmir	1141.155121
15	Gujarat Region	917.8273504
16	Saurashtra & Kutch	496.3982906
17	West Madhya Pradesh	948.1482838
18	Himachal Pradesh	1256.781197

- Spark successfully loaded and processed the rainfall dataset.
- Average annual rainfall was calculated for all 36 subdivisions.
- Coastal Karnataka, Arunachal Pradesh, and Konkan & Goa recorded the highest rainfall.
- Results were generated efficiently using distributed processing.

RESULTS & KEY INSIGHTS

The rainfall prediction and climate analysis system was successfully developed using historical rainfall data from 36 meteorological subdivisions of India. Data preprocessing, exploratory analysis, rainfall categorization, machine learning modeling, future forecasting, rainfall alert generation, and farmer advisory recommendations were completed successfully.

Among the evaluated machine learning models, **XGBoost** achieved the best performance with an **R² score of 0.9872** and a **MAPE of 2.29%**, outperforming Random Forest and becoming the final forecasting model. Future rainfall forecasts were generated for all subdivisions for the period **2018–2030** using historical rainfall patterns, lag features, and rolling averages.

The analysis revealed that the **JJAS (Southwest Monsoon)** season contributes the largest share of annual rainfall across India. Coastal Karnataka, Arunachal Pradesh, Kerala, Konkan & Goa, and Andaman & Nicobar Islands recorded the highest average rainfall, while West Rajasthan and Saurashtra & Kutch showed comparatively lower rainfall levels. Feature importance analysis indicated that historical rainfall trends and rolling rainfall averages were the most influential predictors.

The rainfall alert system generated **277 Normal**, **121 Heavy Rainfall Alert**, and **70 Drought Alert** records during forecast analysis. Based on these alerts, the farmer advisory system provided appropriate recommendations to support agricultural planning and decision-making.

Interactive visualization modules were developed using **Streamlit, Tableau, and Power BI**, while **SQL, R Programming, Hadoop MapReduce, and Apache Spark** were used for database analysis, statistical computing, and distributed data processing. The project successfully integrated machine learning, big data technologies, and visualization tools into a unified rainfall forecasting and climate analysis platform.

CONCLUSION

This project successfully developed a comprehensive **Rainfall Prediction and Climate Analysis System** using historical rainfall data collected from 36 meteorological subdivisions of India. The study involved data preprocessing, exploratory data analysis, rainfall categorization, feature engineering, machine learning model development, future rainfall forecasting, rainfall alert generation, and farmer advisory recommendations.

Among the evaluated models, **XGBoost** achieved the highest prediction accuracy and was selected as the final forecasting model. The system successfully generated rainfall forecasts for the period **2018–2030**, identified potential drought and heavy rainfall conditions, and provided suitable agricultural advisory recommendations based on forecasted rainfall patterns.

In addition, interactive visualization and analysis modules were developed using **Streamlit, Tableau, Power BI, SQL, R Programming, Hadoop MapReduce, and Apache Spark**, enabling effective exploration and interpretation of rainfall data. Overall, the project demonstrates how machine learning, data analytics, and big data technologies can be integrated to support climate monitoring, agricultural planning, water resource management, and informed decision-making.

FUTURE ENHANCEMENTS

The current system provides effective rainfall analysis, forecasting, and decision-support capabilities. However, several enhancements can be incorporated in future versions to improve accuracy, scalability, and real-world applicability.

- Integrate real-time weather and meteorological data from online sources for continuous rainfall monitoring.
- Explore advanced deep learning models such as LSTM, GRU, and Transformer networks for improved rainfall forecasting accuracy.
- Extend the forecasting period and support district-level or city-level rainfall predictions.
- Incorporate additional climate variables such as temperature, humidity, wind speed, and atmospheric pressure to enhance prediction performance.
- Develop a mobile application for real-time rainfall alerts and farmer advisory notifications.
- Integrate Geographic Information Systems (GIS) for spatial rainfall visualization and mapping.
- Implement cloud-based deployment for large-scale data processing and accessibility.
- Enhance the farmer advisory system with crop-specific recommendations based on regional agricultural conditions.
- Develop multilingual support to improve accessibility for farmers across different regions.
- Integrate IoT-based weather sensors for automated data collection and real-time environmental monitoring.

These enhancements can further strengthen the system and make it more useful for climate monitoring, agricultural planning, disaster management, and water resource management.